

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284282

Kind Code

A1

Publication Date

September 11, 2025

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SYSTEMS AND METHODS FOR SIMULTANEOUS LOCALIZATION AND MAPPING IN UNDERGROUND ENVIRONMENTS

Abstract

Systems and computer-implemented methods of simultaneous localization and mapping in underground environments are provided. The method comprises: generating mapping data of the underground environment and inertial measurement data of an agent traversing the underground environment; generating, by the agent, reduced mapping data by performing data reduction of the generated mapping data; updating, by the agent, a local map based on correlated features in the reduced mapping data; updating, by the agent, a localized position of the agent with reference to the updated local map; and providing, by the agent via a user interface, a localization and mapping output including at least one of the updated local map and the updated localized position of the agent.

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Family ID: 93430585

Appl. No.: 18/936637

Filed: November 04, 2024

Related U.S. Application Data

us-provisional-application US 63563738 20240311

Publication Classification

Int. Cl.: G05D1/246 (20240101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/563,738 filed Mar. 11, 2024, and the entire contents of U.S. Provisional Patent Application No. 63/563,738 are hereby incorporated herein in its entirety.

FIELD

[0002] The described embodiments relate to systems and methods for simultaneous localization and mapping, and specifically to systems and methods for simultaneous localization and mapping in underground environments.

BACKGROUND

[0003] Global Navigation Satellite Systems (GNSS) (e.g., Global Positioning System (GPS)) enables satellite-based positioning and navigation systems for many applications. GNSS technology is used, for example, in global shipping, air traffic control, various industrial applications, personal use (e.g., smartphone or other handheld consumer electronics-based applications), military etc. The prevalence of readily available GPS and other GNSS-related technologies has, however, stunted widespread commercial development of high-fidelity positioning and navigation technologies for environments in which line-of-sight satellite communications may not be available. The lack of complementary technologies to GNSS can be an obstacle to growth of industries/applications that require alternatives to GNSS technology. Such industries/applications can include, for example, mining and tunneling applications where positioning knowledge may be critical for worker safety but where positioning and navigation using GNSS is impractical.

SUMMARY

[0004] The following summary is provided to introduce the reader to the more detailed discussion to follow. The summary is not intended to limit or define any claimed or as yet unclaimed invention. One or more inventions may reside in any combination or sub-combination of the elements or process steps disclosed in any part of this document including its claims and figures.

[0005] In a first aspect, there is provided a computer-implemented method of simultaneous localization and mapping in underground environments. The method comprises: generating mapping data of the underground environment and inertial measurement data of an agent traversing the underground environment; generating, by the agent, reduced mapping data by performing data reduction of the generated mapping data; updating, by the agent, a local map based on correlated features in the reduced mapping data; updating, by the agent, a localized position of the agent with reference to the updated local map; and providing, by the agent via a user interface, a localization and mapping output including at least one of the updated local map and the updated localized position of the agent.

[0006] In one or more embodiments, the mapping data includes any combination of range data, bearing data and elevation data of the underground environment with reference to the agent.

[0007] In one or more embodiments, the mapping data includes at least one of i) image data generated by one or more cameras and ii) point cloud data generated by one or more Light Detection and Ranging (LiDAR) sensors.

[0008] In one or more embodiments, the inertial measurement data includes linear acceleration and angular velocity of the agent along three mutually perpendicular axes.

[0009] In one or more embodiments, the reduced mapping data is generated by inputting the mapping data into one or more machine learning models trained to project the mapping data to a

lower dimensional latent space.

[0010] In one or more embodiments, the one or more machine learning models are trained using training data customized for underground mining environments.

[0011] In one or more embodiments, the localization and mapping output further includes an insight or recommendation for improving the simultaneous localization and mapping.

[0012] In a second aspect, there is provided a system for simultaneous localization and mapping in underground environments. The system comprises an agent configured to traverse the underground environment. The agent has an inertial sensor configured to generate inertial measurement data of the agent; a mapping sensor configured to generate mapping data of the underground environment; a memory storing processor-executable instructions; and a processor communicatively coupled to the memory. The instructions configure the processor to: receive the inertial measurement data and the mapping data; generate reduced mapping data by performing data reduction of the received mapping data; update a local map based on correlated features in the reduced mapping data; update a localized position of the agent with reference to the updated local map; and provide, via a user interface, a localization and mapping output including at least one of the updated local map and the updated localized position of the agent.

[0013] In one or more embodiments, at least one of the inertial sensor and the mapping sensor includes multiple sensors collocated on a single substrate.

[0014] In one or more embodiments, the processor is collocated on a same substrate as at least one of the inertial sensor and the mapping sensor.

[0015] In one or more embodiments, the mapping sensor includes one or more cameras, and the mapping data includes image data generated by the one or more cameras.

[0016] In one or more embodiments, the mapping sensor includes one or more Light Detection and Ranging (LiDAR) sensors and the mapping data includes point cloud data generated by the one or more LiDAR sensors.

[0017] In one or more embodiments, the inertial measurement data includes linear acceleration and angular velocity of the agent along three mutually perpendicular axes.

[0018] In one or more embodiments, the processor is configured to generate the reduced mapping data by inputting the mapping data into one or more machine learning models trained to project the mapping data to a lower dimensional latent space.

[0019] In one or more embodiments, the one or more machine learning models are trained using training data customized for underground mining environments.

[0020] In one or more embodiments, the localization and mapping output further includes an insight or recommendation for improving the simultaneous localization and mapping.

[0021] In a third aspect, there is provided a computer-implemented method of simultaneous localization and mapping in underground environments. The method comprises: receiving, by a central server, reduced mapping data and localized position data from each of multiple agents traversing the underground environment; updating, by the central server, a global map by combining the reduced mapping data from the multiple agents and performing cross-correlation across the multiple agents; updating, by the central server, global positioning of each of the multiple agents based on the received localized position data and the updated global map; providing, by the central server, the updated global map and the updated global positioning to each of the multiple agents; and providing, by the central server via a user interface, a localization and mapping output including at least one of the updated global map and the updated global positioning of one or more of the multiple agents.

[0022] In one or more embodiments, the central server is implemented by dynamically assigning central server functionalities dynamically between the multiple agents.

[0023] In one or more embodiments, the method further comprises updating a mapping confidence level associated with a region of the underground environment based on a time duration since previous scan of the region.

[0024] In a fourth aspect, there is provided a system for simultaneous localization and mapping in underground environments. The system comprises multiple agents configured to traverse the underground environment and a central server. Each of the multiple agents has an inertial sensor configured to generate inertial measurement data of the agent and a mapping sensor configured to generate mapping data of the underground environment, each of the multiple agents being configured to generate reduced mapping data and localized position data for that agent based on the mapping data and the inertial measurement data of that agent. The central server has a memory storing processor-executable instructions and a processor communicatively coupled to the memory. The instructions configure the processor to: receive the reduced mapping data and localized position data from each of the multiple agents; update a global map by combining the reduced mapping data from the multiple agents and performing cross-correlation across the multiple agents; update global positioning of each of the multiple agents based on the received localized position data and the updated global map; provide the updated global map and the updated global positioning to each of the multiple agents; and provide, via a user interface, a localization and mapping output including at least one of the updated global map and the updated global positioning of one or more of the multiple agents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The drawings included herewith are for illustrating various examples of systems, methods, and devices of the teaching of the present specification and are not intended to limit the scope of what is taught in any way.

[0026] FIG. 1 is a schematic diagram showing a system for simultaneous localization and mapping, in accordance with one or more embodiments.

[0027] FIG. 2 is a block diagram showing components of an example agent of the system of FIG. 1.

[0028] FIG. 3 is a flowchart showing a computer-implemented method of simultaneous localization and mapping, in accordance with one or more embodiments.

[0029] FIG. 4 is a schematic representation of an example data reduction of mapping data performing during the method of FIG. 3.

[0030] FIG. 5 is a schematic representation of an example GUI for providing a localization and mapping output of the method of FIG. 3.

[0031] FIG. 6 is a schematic diagram showing another system for simultaneous localization and mapping, in accordance with one or more embodiments.

[0032] FIG. 7 is a block diagram showing components of an example central server of the system of FIG. 6.

[0033] FIG. 8 is a flowchart showing another computer-implemented method of simultaneous localization and mapping, in accordance with one or more embodiments.

DETAILED DESCRIPTION

[0034] Several example embodiments are described herein. It will be appreciated that numerous specific details are set forth in order to provide a thorough understanding of the example embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures and components have not been described in detail so as not to obscure the embodiments described herein. Furthermore, this description and the drawings are not to be considered as limiting the scope of the embodiments described herein in any way, but rather as merely describing the implementation of the various embodiments described herein.

[0035] It should be noted that terms of degree such as “substantially”, “about” and “approximately”

when used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed. These terms of degree should be construed as including a deviation of the modified term if this deviation would not negate the meaning of the term it modifies.

[0036] In addition, as used herein, the wording “and/or” is intended to represent an inclusive-or. That is, “X and/or Y” is intended to mean X or Y or both, for example. As a further example, “X, Y, and/or Z” is intended to mean X or Y or Z or any combination thereof.

[0037] The embodiments of the systems and methods described herein may be implemented in hardware or software, or a combination of both. These embodiments may be implemented in computer programs executing on programmable computers, each computer including at least one processor, a data storage system (including volatile memory or non-volatile memory or other data storage elements or a combination thereof), and at least one communication interface. For example and without limitation, the programmable computers (referred to below as computing devices) may be a server, network appliance, embedded device, computer expansion module, personal computer, laptop, personal data assistant, cellular telephone, smart-phone device, tablet computer, wireless device or any other computing device capable of being configured to carry out the methods described herein.

[0038] In some embodiments, the communication interface may be a network communication interface. In embodiments in which elements are combined, the communication interface may be a software communication interface, such as those for inter-process communication (IPC). In still other embodiments, there may be a combination of communication interfaces implemented as hardware, software, and a combination thereof.

[0039] Program code may be applied to input data to perform the functions described herein and to generate output information. The output information is applied to one or more output devices, in known fashion.

[0040] Each program may be implemented in a high-level procedural or object-oriented programming and/or scripting language, or both, to communicate with a computer system. However, the programs may be implemented in assembly or machine language, if desired. In any case, the language may be a compiled or interpreted language. Each such computer program may be stored on a storage media or a device (e.g. ROM, magnetic disk, optical disc) readable by a general or special purpose programmable computer, for configuring and operating the computer when the storage media or device is read by the computer to perform the procedures described herein. Embodiments of the system may also be considered to be implemented as a non-transitory computer-readable storage medium, configured with a computer program, where the storage medium so configured causes a computer to operate in a specific and predefined manner to perform the functions described herein.

[0041] Furthermore, the systems, processes and methods of the described embodiments are capable of being distributed in a computer program product comprising a computer readable medium that bears computer usable instructions for one or more processors. The medium may be provided in various forms, including one or more diskettes, compact disks, tapes, chips, wireline transmissions, satellite transmissions, internet transmission or downloads, magnetic and electronic storage media, digital and analog signals, and the like. The computer useable instructions may also be in various forms, including compiled and non-compiled code.

[0042] Various embodiments have been described herein by way of example only. Various modifications and variations may be made to these example embodiments without departing from the spirit and scope of the invention, which is limited only by the appended claims. Also, in the various user interfaces illustrated in the figures, it will be understood that the illustrated user interface text and controls are provided as examples only and are not meant to be limiting. Other suitable user interface elements may be possible.

[0043] The disclosed systems and methods can enable simultaneous mapping and localization in

underground environments in which line-of-sight satellite-based communication may not be available. For example, the disclosed systems and methods may be used in mining and tunneling applications. The disclosed system and methods can enable the shift of mining and tunneling towards electrification to improve reliability, efficiency, and workplace safety. The disclosed system and methods can provide real-time, dynamic, and collaborative mapping and positioning for vehicles, personnel, and assets while informing users about changing conditions and environmental hazards on site. The disclosed system and methods can provide a robust underground positioning system to enable a transition towards fully autonomous mines.

[0044] In some applications, the disclosed systems and methods may be used to complement GNSS technology to provide improvement in accuracy, precision, and/or reliability. For example, the localization and mapping data generated using the disclosed systems and methods may be used to complement GNSS data in critical infrastructure and military applications. GNSS technology (e.g., GPS) may lack systematic redundancy required in critical infrastructure and military applications. The disclosed systems and methods can provide redundancy in such applications.

[0045] The disclosed systems and methods can serve as a robust replacement for geo-positioning systems for applications that are unable to tolerate single-point systematic failure. The disclosed systems and methods can provide a completely independent means for high-precision positioning for civilian and military applications in which systems like GPS may be intermittently or actively blocked.

[0046] In some applications, the disclosed systems and methods may be used in combination with other alternatives to GNSS-based navigation technologies, for example, Celestial-Aided Navigation, Magnetic Anomaly-Aided Navigation, and Vision-Aided Navigation. The disclosed systems and methods can provide reliable reference points to calibrate and utilize these alternative technologies with greater precision and accuracy.

[0047] Reference is first made to FIG. 1. FIG. 1 is a schematic diagram showing an example system **100** for simultaneous localization and mapping in an environment **20**. Environment **20** can be an underground environment (e.g., a mine or a tunnel) where line-of-sight satellite communication may not be available for mapping and positioning using GNSS technology.

[0048] System **100** includes an agent **110**. Agent **110** can be any suitable device that traverses and provides sensor data for environment **20**. For example, agent **110** may include a mobile user device or a device mounted on a vehicle (e.g., a mining vehicle, an unmanned aerial vehicle etc.). In the illustrated example, agent **110** traverses along path **120** and captures sensor data for various portions **30** (e.g., portions **30a** and **30b**) of environment **20**.

[0049] In some embodiments, agent **110** may traverse along all three dimensions of a three-dimensional environment **20**. In other embodiments, agent **110** may only traverse along one or two dimensions of a three-dimensional environment **20**.

[0050] Reference is now made to FIGS. 1 and 2. FIG. 2 is a block diagram showing components of agent **110**. In the illustrated example, agent **110** includes an inertial sensor **210**, a mapping sensor **220**, a communication unit **230**, a memory unit **240**, a processor unit **250**, and a power unit **260**. In other examples, agent **110** may include fewer or greater number of components. For example, agent **110** may include multiple inertial sensors **210** and/or multiple mapping sensors **220**.

[0051] Inertial sensor **210** may include any suitable combination of number and types of sensors configured to generate inertial measurement data for agent **110**. For example, inertial sensor **210** may include one or more sensors configured to generate inertial measurement data that includes linear acceleration and angular velocity of agent **110** along three mutually perpendicular axes. In some embodiments, inertial sensor **210** may include multiple sensors collocated on a single substrate. Multiple sensors collocated on a single substrate may enable a more compact design of agent **110**. In other embodiments, inertial sensor **210** may include multiple sensors located on different substrates. In some examples, the relative location of each of the multiple sensors with respect to the other sensors of inertial sensor **210** may be fixed. In other examples, the relative

location of at least one of the multiple sensors with respect to the other sensors of inertial sensor **210** may vary with time. Multiple sensors collocated on different substrates may provide greater modularity and/or design flexibility.

[0052] Mapping sensor **220** may include any suitable combination of number and types of sensors configured to generate mapping data of environment **20**. The mapping data may include, for example, range, bearing and/or elevation data of environment **20** with reference to agent **110**. In some embodiments, mapping sensor **220** may include one or more cameras configured to generate image data of environment **20**. The image data may include monochromatic image data. Optionally, the image data may include multiple datasets of monochromatic image data where the multiple datasets correspond to different wavelengths. In some embodiments, mapping sensor **220** may include one or more Light Detection and Ranging (LiDAR) sensors configured to generate point cloud data of environment **20** with reference to agent **110**. The point cloud data may include three-dimensional point cloud data.

[0053] In some embodiments, mapping sensor **220** may include multiple sensors collocated on a single substrate. Optionally, mapping sensor **220** and inertial sensor **210** may be located on the same substrate. Multiple sensors collocated on a single substrate may enable a more compact design of agent **110**. In other embodiments, mapping sensor **220** may include multiple sensors located on different substrates. In some examples, the relative location of each of the multiple sensors with reference to the other sensors of mapping sensor **220** may be fixed. In other examples, the relative location of at least one of the multiple sensors with reference to the other sensors of mapping sensor **220** may vary with time. Multiple sensors collocated on different substrates may provide greater modularity and/or design flexibility. In some embodiments, a single sensor may provide combined functionalities of inertial sensor **210** and mapping sensor **220**.

[0054] Communication unit **230** may include any suitable design to provide bidirectional communication for agent **110** with other devices or computers. Communication unit **230** may include wired or wireless connection capabilities. Communication unit **230** may enable the bidirectional communication using any suitable communication network. The communication network may include, for example, the Internet, a Wide-Area Network (WAN), a Local-Area Network (LAN), a point-to-point connection, etc. The communication may be implemented using time-stamped communication messages. In some embodiments, communication unit **230** may enable low-latency communication for agent **110**.

[0055] Power unit **260** may have any suitable design to provide power to other components of agent **110**. For example, power unit **260** may include a battery pack.

[0056] Processor unit **250** may have any suitable design to perform operations of agent **110**. Processor unit **250** can be any suitable processor, controller or digital signal processor that can provide sufficient processing power depending on the configuration, purposes and requirements of agent **110**, as is known by those skilled in the art. For example, processor unit **250** may be a high-performance general processor. As another example, processor unit **250** may include a standard processor, such as an Intel® processor, or an AMD® processor. As another example, processor unit **250** can include more than one processor with each processor being configured to perform different dedicated tasks. As another example, specialized hardware (e.g., graphical processing units (GPUs)) can be used provide some of the functions provided by processor unit **250**.

[0057] Processor unit **250** may control the operation of other components of agent **110**. In some embodiments, processor unit **250** may determine position, motion and/or orientation of agent **110** with reference to a local map for one or more time periods. Processor unit **250** may determine the position, motion and/or orientation based on inertial measurement data generated by inertial sensor **210** and/or mapping data generated by mapping sensor **220**.

[0058] In some embodiments, processor unit **250** may be collocated on the same substrate as inertial sensor **210** and/or mapping sensor **220**. This may enable greater communication efficiency between processor unit **250** and the sensor(s). In some embodiments, processor unit **250** may be at

a remote location and communicate with agent **110** using a communication network. This may enable a more compact design for agent **110**.

[0059] Memory unit **240** can include software code for implementing an operating system **270**, programs **280** and a database **290**. Memory unit **240** can include RAM, ROM, one or more hard drives, one or more flash drives or some other suitable data storage elements such as disk drives, etc. In some embodiments, memory unit **240** may be at a remote location and communicate with agent **110** using a communication network. This may enable a more compact design for agent **110**. In some embodiments, memory unit **240** may be integrated into processor unit **250**. This may enable faster memory read/write operations for processor unit **250**.

[0060] Memory unit **240** can be used to store an operating system **270** and programs **280**, as is commonly known by those skilled in the art. For instance, operating system **270** provides various basic operational processes for agent **110**. For example, the operating system **270** may be an operating system such as Windows® Server operating system, or Red Hat® Enterprise Linux (RHEL) operating system, or another operating system. Programs **280** can include various programs so that agent **110** can perform various functions such as, but not limited to, receiving sensor data, performing simultaneous localization and mapping, etc.

[0061] Database **290** may include a Structured Query Language (SQL) database such as PostgreSQL or MySQL or a not only SQL (NoSQL) database such as MongoDB, or Graph Databases, etc. Database **290** may store inertial measurement data received from inertial sensor **210** and/or mapping data received from mapping sensor **220**. In some embodiments, database **290** may store position, motion and/or orientation data of agent **110** for one or more time periods. Database **290** may store one or more maps of environment **20**. For example, the map may be an initial map received from an external device. As another example, database **290** may store multiple time-stamped maps of environment **20** generated by processor unit **250**. In some embodiments, database **290** may store one or more machine learning models used by agent **110**. For example, agent **110** may use the machine learning models to perform data reduction of point cloud data of environment **20**.

[0062] Reference is now made to FIG. **3**. FIG. **3** is a flowchart showing a computer-implemented method **300** of simultaneous localization and mapping. Any suitable system may be used to implement method **300**. For example, method **300** may be implemented using system **100** (FIG. **1**) for simultaneous localization and mapping in environment **20** (FIG. **1**) and concurrent reference is made now to FIGS. **1** to **3**.

[0063] Method **300** may be performed at various times. For example, method **300** may be performed based on an automated schedule (e.g. a fixed time interval, a maximum period after previous mapping, etc.). As another example, method **300** may be continuously performed. As another example, method **300** may be automatically performed when a control input is provided (e.g., by a system user) to system **100**.

[0064] At act **310**, method **300** includes generating mapping data and inertial measurement data. For example, inertial sensor **210** can generate inertial measurement data that includes linear acceleration and angular velocity values of agent **110** along three mutually perpendicular axes. As another example, mapping sensor **220** can generate mapping data including, for example, range, bearing and/or elevation data of environment **20** with reference to agent **110**. The mapping data may include image data and/or point cloud data.

[0065] At act **320**, method **300** includes generating reduced mapping data by performing data reduction of the mapping data generated at act **310**. For example, processor unit **250** may generate reduced mapping data at act **320** by performing data reduction of the generated mapping data.

[0066] Reference is now additionally made to FIG. **4** showing a schematic representation of an example data reduction of mapping data. FIG. **4** shows point cloud data **435** in raw data space **430** captured by an agent at a first capture time **410** and a subsequent second capture time **420** following agent motion **425**. FIG. **4** illustrates an example that includes point cloud data. In other

examples, the capture data may be image data and/or point cloud data. FIG. 4 illustrates difficulties associated (e.g., required computational complexity) with correlation of the captured data for the two capture times **410** and **420**. As the number of data points per capture time included in the point cloud data **435** increases (e.g., on the order of millions of data points), the required computational resources for analysis can be impracticably high.

[0067] Method **300** can enable reduction in computational resources required for correlation of the captured data by generating the reduced mapping data. Processor unit **250** can use one or more trained machine learning models to project raw captured mapping data to a lower dimensional latent space **450**. FIG. 4 illustrates a simplified example of the data reduction to recognizable geometric shapes **455**. In other examples, the parameterized space **450** may be specifically tailored to fully leverage the scale and abstract topology that best describes environment **20**. For example, the machine learning models may be trained using application-specific training data (e.g., specific training data for underground mining environments).

[0068] At act **330**, method **300** includes updating a local map based on correlated or matched features in the reduced mapping data. The local map may be a local map that was previously updated using method **300**. In some embodiments, system **100** may receive an initial local map from an external device. The reduced mapping data can enable identification of matched features **455** between the first capture time **410** and the second capture time **420** with reduced computation resources. The matched features **455** and/or the unmatched features **460** can provide mapping information of the environment. The mapping information can be used to update a local map **480** of the environment **20**.

[0069] At act **340**, method **300** includes updating a localized position of agent **110** with reference to the updated local map. The localized position may be a localized position that was previously updated using method **300**. If no previously updated localized position exists, an initial localized position may be generated at act **340**. Processor unit **250** may update the localized position by determining a position, motion and/or orientation of agent **110** over a time period based on the inertial measurement data and the reduced mapping data. Processor unit **250** can determine motion of agent **110** during the time period between first capture time **410** and second capture time **420** based on the identified matched features **155**. In some embodiments, processor unit **250** may determine three-dimensional position, speed and/or acceleration of agent **110** with reference to the updated local map for a time period. In some embodiments, processor unit **250** may determine three-dimensional orientation of agent **110** with reference to the updated local map for a time period. For example, processor unit **250** may determine yaw, pitch and roll rate of agent **110** over a time period.

[0070] At act **350**, method **300** includes providing a localization and mapping output. The localization and mapping output may be provided via a user interface. In some embodiments, the localization and mapping output may be provided via an output file or may be stored in a database (e.g., database **290**).

[0071] The localization and mapping output may include the updated local map of environment **20** and/or localized position information of agent **110** in near-real-time for user access and/or for higher-level data processing. In some embodiments, the localization and mapping output may include insights and/or recommendations (e.g., insights related to localization and mapping performance of agent **110**, metadata related to early warning events, recommendations to improve map accuracy including, for example, agent positioning to map a low confidence region, etc.). One or more machine learning models may be used to generate the insights and/or recommendations.

[0072] Reference is now additionally made to FIG. 5. FIG. 5 shows an example GUI **500** for providing a localization and mapping output of an underground environment **20** to a user. GUI **500** may include multiple GUI elements including, for example, a pane, 3D graphics display, a console and/or a menu bar. The relative positioning of the GUI elements may be reconfigurable by a user. GUI **500** can enable a user to visualize and/or interact with agent **110** and/or assets (e.g., personnel,

equipment, etc.) in underground environment **20**. GUI **500** can display continuously updated positioning for agent **110** in underground environment **20** and/or enable continuous tracking of assets within underground environment **20**.

[0073] Reference is now made to FIG. **6**. FIG. **6** is a schematic diagram showing an example system **600** for simultaneous localization and mapping in an environment **40**. In the illustrated example, system **600** includes multiple agents **110** and a central server **605**. In other examples, system **600** may not include a dedicated central server. An agent **110** of the multiple agents **110** may provide the functionalities of the central server. Optionally, the central server functionalities may be dynamically assigned between different agents **110**. This may enable higher efficiency utilization of available computational resources at agents **110**. In some embodiments, central server **605** may be implemented as multiple servers using distributed computing. This may enable higher efficiency utilization of available computational resources.

[0074] Environment **40** can be an underground environment (e.g., a mine or a tunnel) where line-of-sight satellite communication may not be available for mapping and positioning using GNSS technology. In some embodiments, central server **605** may be located within environment **40**. In other embodiments, central server **605** may be remotely located.

[0075] The multiple agents of system **600** may simultaneously traverse environment **40** generating mapping data and inertial measurement data. Each agent **110** may traverse along different paths **640** and capture data for various portions **620** and **630** of environment **40**. In some embodiments, each agent **110** may traverse along all three dimensions of a three-dimensional environment **40**. In other embodiments, each agent **110** may only traverse along one or two dimensions of a three-dimensional environment **40**. Environment **40** may be divided into one or more zones. In some embodiments, each agent **110** may be assigned to specific zone(s). This may improve mapping efficiency of environment **40** by limiting the overlapping area mapped by multiple agents.

[0076] In some embodiments, each of multiple agents **110** may include identical type, number and configuration of inertial and mapping sensors. In other embodiments, different agents **110** may include different type, number and/or configuration of inertial and/or mapping sensors.

[0077] Central server **605** can combine mapping and positioning data generated by each of the multiple agents **110** of system **600**. Central server **605** can generate/update a global map based on the combined data from the multiple agents **110**. The global map may be defined with respect to “absolute” Earth-centered coordinates (e.g., WGS84, latitude and longitude).

[0078] Central server **605** can use potential mapping correlations between overlapping regions scanned by two or more agents **110** for a given time period to improve mapping accuracy of the global map (compared with local maps at each agent **110**) and positioning accuracy of the agents **110** within the global map.

[0079] As the multiple agents **110** traverse environment **40**, different regions of environment **40** may be scanned at different instances in time. Central server **605** may retain mapping information of previously scanned regions as the global map is continuously updated based on combined data generated by multiple agents **110**. A mapping confidence level may be associated with each previously scanned region. For example, a previously scanned region **660** that is not being currently scanned by any of the multiple agents **110** at the time instance illustrated in FIG. **6** may have a high mapping confidence level based on one or more recent previous scans. In some embodiments, central server **605** may update the mapping confidence level of a region based on the time duration after previous scan. System **600** may, for example, perform a larger reduction in the mapping confidence level of a region **670** having a larger time duration after previous scan.

[0080] Reference is now made to FIGS. **6** and **7**. FIG. **7** is a block diagram showing components of central server **605**. In the illustrated example, central server **605** includes an I/O unit **710**, a display **720**, a communication unit **730**, a memory unit **740**, a processor unit **750**, and a power unit **760**. In other examples, central server **605** may include fewer or greater number of components.

[0081] I/O unit **710** can include at least one of a mouse, a keyboard, a touch screen, a thumbwheel,

a trackpad, a trackball, a card-reader, voice recognition software and the like, depending on the particular implementation of central server **605**. In some embodiments, some of these components can be integrated with one another. I/O unit **710** can enable a user to interact with central server **605**.

[0082] Display **720** may be a LED or LCD based display and may be a touch sensitive screen that supports gestures. Display **720** may be integrated into central server **605**. Alternatively, display **720** may be located physically remote from central server **605** and communicate with central server **605** using a communication network. In some embodiments, central server **605** may not include a dedicated display **720**.

[0083] Communication unit **730** may include any suitable design to provide bidirectional communication for central server **605** with other devices or computers (for example, with multiple agents **110**). Communication unit **730** may include wired or wireless connection capabilities. Communication unit **730** may enable the bidirectional communication using any suitable communication network. The communication network may include, for example, the Internet, a Wide-Area Network (WAN), a Local-Area Network (LAN), a point-to-point connection, etc. The communication may be implemented using time-stamped communication messages. In some embodiments, communication unit **730** may enable low-latency communication for central server **605**. Communication unit **730** can enable central server **605** to receive reduced mapping data and localized position data from multiple agents **110**. Communication unit **730** can enable central server **605** to provide global map updates and global positioning updates to each of the multiple agents **110**.

[0084] Power unit **760** may have any suitable design to provide power to other components of central server **605**. For example, power unit **760** may include a battery pack.

[0085] Processor unit **750** may have any suitable design to perform operations of central server **605**. Processor unit **750** can be any suitable processor, controller or digital signal processor that can provide sufficient processing power depending on the configuration, purposes and requirements of central server **605**, as is known by those skilled in the art. For example, processor unit **750** may be a high-performance general processor. As another example, processor unit **750** may include a standard processor, such as an Intel® processor, or an AMD® processor. As another example, processor unit **750** can include more than one processor with each processor being configured to perform different dedicated tasks. As another example, specialized hardware (e.g., graphical processing units (GPUs)) can be used provide some of the functions provided by processor unit **750**.

[0086] Processor unit **750** may control the operation of other components of central server **605**. In some embodiments, processor unit **750** may combine received data from multiple agents to update a global map. Processor unit **750** may update global positioning data of each of the multiple agents based on the received localized position data and the updates to the global map.

[0087] In some embodiments, processor unit **750** may be at a remote location and communicate with central server **605** using a communication network. This may enable a more compact design for central server **605**.

[0088] Memory unit **740** can include software code for implementing an operating system **770**, programs **780** and a database **790**. Memory unit **740** can include RAM, ROM, one or more hard drives, one or more flash drives or some other suitable data storage elements such as disk drives, etc. In some embodiments, memory unit **740** may be at a remote location and communicate with central server **605** using a communication network. This may enable a more compact design for central server **605**. In some embodiments, memory unit **740** may be integrated into processor unit **750**. This may enable faster memory read/write operations for processor unit **750**.

[0089] Memory unit **740** can be used to store an operating system **770** and programs **780**, as is commonly known by those skilled in the art. For instance, operating system **770** provides various basic operational processes for central server **605**. For example, the operating system **770** may be

an operating system such as Windows® Server operating system, or Red Hat® Enterprise Linux (RHEL) operating system, or another operating system. Programs **780** can include various programs so that central server **605** can perform various functions such as, but not limited to, receiving data from multiple agents, performing updates to global maps and global positioning of the multiple agents, providing global map updates and positioning updates to the multiple agents, etc.

[0090] Database **790** may include a Structured Query Language (SQL) database such as PostgreSQL or MySQL or a not only SQL (NoSQL) database such as MongoDB, or Graph Databases, etc. Database **790** may store reduced mapping data and localized position data received from multiple agents **110**. In some embodiments, database **790** may store global map and global positioning data of multiple agents **110** for one or more time periods. Database **790** may store one or more maps of environment **40**. For example, the map may be an initial global map received from an external device. As another example, database **790** may store multiple time-stamped global maps of environment **40**. In some embodiments, database **790** may store one or more machine learning models used by central server **605**. For example, central server **605** may use the machine learning models to perform mapping cross-correlations between multiple agents. In some embodiments, database **790** includes a map features repository that stores historical matched map features. The stored matched map features may be used to perform mapping cross-correlations between multiple agents.

[0091] Reference is now made to FIG. **8**. FIG. **8** is a flowchart showing a computer-implemented method **800** of simultaneous localization and mapping. Any suitable system may be used to implement method **800**. For example, method **800** may be implemented using system **600** (FIG. **6**) for simultaneous localization and mapping in environment **40** (FIG. **6**) and concurrent reference is made now to FIGS. **6** and **7**.

[0092] Method **800** may be performed at various times. For example, method **800** may be performed based on an automated schedule (e.g. a fixed time interval, a maximum period after previous mapping, etc.). As another example, method **800** may be continuously performed. As another example, method **800** may be automatically performed when a control input is provided (e.g., by a system user) to system **600**.

[0093] At act **810**, method **800** includes receiving reduced mapping data and localized position data from multiple agents. For example, central server **605** may receive reduced mapping data and localized position data from each of the multiple agents **110**. Each multiple agent **110** may provide the reduced mapping data and localized position data by executing method **300** (FIG. **3**).

[0094] At act **820**, method **800** includes combining the received reduced mapping data from multiple agents and performing cross-correlation across the multiple agents to update a global map. For example, central server **605** may combine the reduced mapping data received at act **810** and perform cross-correlation across the multiple agents. Mapping correlations between overlapping regions (e.g., region **630**) scanned by two or more agents **110** may enable the global map to have higher accuracy compared with local maps generated at each agent **110**.

[0095] The global map may be a global map that was previously updated using method **800**. In some embodiments, system **600** may receive an initial global map from an external device.

[0096] Central server **605** may use one or more machine learning models to perform cross-correlation across the multiple agents. The machine learning models may be trained to enable identification of matched map features. In some embodiments, central server **605** may use historical matched map features (e.g., stored in a map features repository in database **790**) while performing cross-correlation across the multiple agents.

[0097] At act **830**, method **800** includes updating global positioning of each of the multiple agents based on the updates to the global map and the received localized position data from the multiple agents. For example, central server **605** may update global positioning of each of the multiple agents **110** based on the localized position data received from the agent **110** and the global map

updates from act **820**. The combination of data from multiple agents and the higher accuracy of the global map can enable higher accuracy of the global positioning of agents **110** compared with the localized positioning determined at individual agents **110**.

[0098] At act **840**, method **800** includes providing the global map updates and global positioning updates to each of the multiple agents. For example, central server **605** may provide the global map updates and global positioning updates to each of the multiple agents **110**. Each agent **110** may perform real-time updates to the local map and localized position of that agent **110** based on the data received from central server **605**.

[0099] At act **850**, method **800** includes providing a localization and mapping output. The localization and mapping output may be provided via a user interface. In some embodiments, the localization and mapping output may be provided via an output file or may be stored in a database (e.g., database **290**).

[0100] The localization and mapping output may include the updated global map of environment **40** and/or updated global position information of the multiple agents **110** in near-real-time for user access and/or for higher-level data processing. In some embodiments, the localization and mapping output may include insights and/or recommendations (e.g., insights related to localization and mapping performance of the multiple agents **110**, metadata related to early warning events, recommendations to improve global map accuracy including, for example, agent positioning to map a low mapping confidence region, etc.). One or more machine learning models may be used to generate the insights and/or recommendations.

[0101] The present invention has been described here by way of example only. Various modification and variations may be made to these exemplary embodiments without departing from the spirit and scope of the invention, which is limited only by the appended claims.

Claims

1. A computer-implemented method of simultaneous localization and mapping in underground environments, the method comprising: generating mapping data of the underground environment and inertial measurement data of an agent traversing the underground environment; generating, by the agent, reduced mapping data by performing data reduction of the generated mapping data; updating, by the agent, a local map based on correlated features in the reduced mapping data; updating, by the agent, a localized position of the agent with reference to the updated local map; and providing, by the agent via a user interface, a localization and mapping output including at least one of the updated local map and the updated localized position of the agent.
2. The method of claim 1, wherein the mapping data includes any combination of range data, bearing data and elevation data of the underground environment with reference to the agent.
3. The method of claim 1, wherein the mapping data includes at least one of i) image data generated by one or more cameras and ii) point cloud data generated by one or more Light Detection and Ranging (LiDAR) sensors.
4. The method of claim 1, wherein the inertial measurement data includes linear acceleration and angular velocity of the agent along three mutually perpendicular axes.
5. The method of claim 1, wherein the reduced mapping data is generated by inputting the mapping data into one or more machine learning models trained to project the mapping data to a lower dimensional latent space.
6. The method of claim 5, wherein the one or more machine learning models are trained using training data customized for underground mining environments.
7. The method of claim 1, wherein the localization and mapping output further includes an insight or recommendation for improving the simultaneous localization and mapping.
8. A system for simultaneous localization and mapping in underground environments, the system comprising an agent configured to traverse the underground environment, the agent having: an

inertial sensor configured to generate inertial measurement data of the agent; a mapping sensor configured to generate mapping data of the underground environment; a memory storing processor-executable instructions; and a processor communicatively coupled to the memory, the instructions configuring the processor to: receive the inertial measurement data and the mapping data; generate reduced mapping data by performing data reduction of the received mapping data; update a local map based on correlated features in the reduced mapping data; update a localized position of the agent with reference to the updated local map; and provide, via a user interface, a localization and mapping output including at least one of the updated local map and the updated localized position of the agent.

9. The system of claim 8, wherein at least one of the inertial sensor and the mapping sensor includes multiple sensors collocated on a single substrate.

10. The system of claim 8, wherein the processor is collocated on a same substrate as at least one of the inertial sensor and the mapping sensor.

11. The system of claim 8, wherein the mapping sensor includes one or more cameras, and the mapping data includes image data generated by the one or more cameras.

12. The system of claim 8, wherein the mapping sensor includes one or more Light Detection and Ranging (LiDAR) sensors and the mapping data includes point cloud data generated by the one or more LiDAR sensors.

13. The system of claim 8, wherein the inertial measurement data includes linear acceleration and angular velocity of the agent along three mutually perpendicular axes.

14. The system of claim 8, wherein the processor is configured to generate the reduced mapping data by inputting the mapping data into one or more machine learning models trained to project the mapping data to a lower dimensional latent space.

15. The system of claim 14, wherein the one or more machine learning models are trained using training data customized for underground mining environments.

16. The system of claim 8, wherein the localization and mapping output further includes an insight or recommendation for improving the simultaneous localization and mapping.

17. A computer-implemented method of simultaneous localization and mapping in underground environments, the method comprising: receiving, by a central server, reduced mapping data and localized position data from each of multiple agents traversing the underground environment; updating, by the central server, a global map by combining the reduced mapping data from the multiple agents and performing cross-correlation across the multiple agents; updating, by the central server, global positioning of each of the multiple agents based on the received localized position data and the updated global map; providing, by the central server, the updated global map and the updated global positioning to each of the multiple agents; and providing, by the central server via a user interface, a localization and mapping output including at least one of the updated global map and the updated global positioning of one or more of the multiple agents.

18. The method of claim 17, wherein the central server is implemented by dynamically assigning central server functionalities dynamically between the multiple agents.

19. The method of claim 17, further comprising updating a mapping confidence level associated with a region of the underground environment based on a time duration since previous scan of the region.
